

# Park, Seonghoon

Postdoc Associate at Virginia Tech | Ph.D. in Computer Science

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## RESEARCH INTERESTS

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### On-device and embedded artificial intelligence

As mobile applications increasingly employ deep neural networks (DNNs), efficient and accurate execution on resource-constrained devices such as embedded devices and mobile devices has become critical. My research focuses on enabling such efficiency across various DNN tasks, including vision foundation models [c8], super-resolution for 360-degree videos [c6, c11], vision transformers, and real-time gaze tracking [c3].

### Energy-aware mobile systems

Reducing energy consumption has consistently been a crucial concern for mobile devices. I have researched energy optimization for native [c1], web [c2, c9], and game applications [j1] on smartphones. I am also interested in energy-efficient on-device AI techniques [j2, c11] and AI-based energy optimization strategies [c2, c9].

### Mobile immersive computing

Immersive videos, such as omnidirectional and volumetric videos, can provide interactive and engaging experiences on mobile devices, but their large data sizes and computational demands pose significant technical challenges. I have explored techniques to maximize video quality under time and resource constraints, such as super-resolution for live 360-degree videos [c6, c11] and 3DGS-based volumetric video streaming [c10].

### Cross-device computing

As users increasingly interact with multiple personal devices, cross-device computing has garnered interest. However, prior approaches often suffer from platform dependency. My research explores platform-agnostic methods for I/O sharing and interface distribution by leveraging the meta-platform characteristics of the web. I have analyzed user experience on the mobile web [c4] and proposed cross-device web computing techniques [c5, c7].

## EXPERIENCES

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### Postdoctoral Associate

Oct. 2025 – Present

Virginia Tech, VA, USA

Supervisors:

- Prof. Lingjia Liu (Department of Electrical and Computer Engineering)
- Prof. Bo Ji (Department of Computer Science)

## EDUCATION

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### Ph.D. in Computer Science

Mar. 2018 – Aug. 2025

Yonsei University, Seoul, Republic of Korea

- Advisor: Prof. Hojung Cha (Mobile Embedded Systems Lab.)
- Thesis: Providing Resource-Optimized User Experiences in Networked Mobile Systems

### B.S in Computer Science and Engineering

Mar. 2014 – Feb. 2018

Yonsei University, Seoul, Republic of Korea

## PUBLICATIONS (PEER-REVIEWED)

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'BK21 IF' refers to the IF listed for top CS conferences by the National Research Foundation of Korea.

\* Co-primary authors

### Conference Papers

[c11] EOS: Energy-Optimized Super-Resolution on Mobile Devices for Live 360-Degree Videos

[Seonghoon Park](#), Minchan Kim, Hyejin Park, Jeho Lee, Jiwon Kim, and Hojung Cha

The 31st Annual International Conference on Mobile Computing and Networking

[ACM MobiCom 2025](#). November 4–8, 2025. Hong Kong, China

(Acceptance rate: 10.3% for the winter round; BK21 IF=4; Top conference in mobile computing)

[c10] Vega: Fully Immersive Mobile Volumetric Video Streaming with 3D Gaussian Splatting

Gunjoong Kim\*, [Seonghoon Park](#)\*, Jeho Lee, Chanyoung Jung, Hyungchol Jun, and Hojung Cha

The 31st Annual International Conference on Mobile Computing and Networking

[ACM MobiCom 2025](#). November 4–8, 2025. Hong Kong, China

(Acceptance rate: 10.3% for the winter round; BK21 IF=4; Top conference in mobile computing)

[c9] Ember: Task Wakeup Sequence–Based Energy Optimization for Mobile Web Browsing

[Seonghoon Park](#), Jiwon Kim, Jeho Lee, and Hojung Cha

The ACM SIGBED International Conference on Embedded Software

[ACM EMSOFT 2025](#). September 28–October 3, 2025. Taipei, Taiwan

(Acceptance rate: 27%, BK21 IF=2; Top conference in embedded systems)

Also published in ACM Transactions on Embedded Computing Systems (JCR 2024 IF=2.6, Q2)

[c8] ARIA: Optimizing Vision Foundation Model Inference on Heterogeneous Mobile Processors for Augmented Reality

Chanyoung Jung\*, Jeho Lee\*, Gunjoong Kim, Jiwon Kim, [Seonghoon Park](#), and Hojung Cha

The 23rd Annual International Conference on Mobile Systems, Applications and Services

[ACM MobiSys 2025](#). June 23–27, 2025. Anaheim, California, US.

(Acceptance rate: 18.0%; BK21 IF=3; Top conference in mobile systems; **Best paper award!**)

[c7] Vulture: Cross-Device Web Experience with Fine-Grained Graphical User Interface Distribution

[Seonghoon Park](#), Jeho Lee, Yonghun Choi, and Hojung Cha

IEEE INFOCOM 2024 – IEEE Conference on Computer Communications

[IEEE INFOCOM 2024](#). May 20–23, 2024. Vancouver, Canada.

(Acceptance rate: 19.6%; BK21 IF=4; Top conference in computer networks)

- [c6] OmniLive: Super-Resolution Enhanced 360° Video Live Streaming for Mobile Devices  
[Seonghoon Park](#)\*, Yeonwoo Cho\*, Hyungchol Jun, Jeho Lee, and Hojung Cha  
The 21st Annual International Conference on Mobile Systems, Applications and Services  
ACM MobiSys 2023. June 18–22, 2023. Helsinki, Finland.  
(Acceptance rate: 20.7%; BK21 IF=3; Top conference in mobile systems)
- [c5] Crow API: Cross-device I/O Sharing in Web Applications  
[Seonghoon Park](#), Jeho Lee, and Hojung Cha  
IEEE INFOCOM 2023 – IEEE Conference on Computer Communications  
IEEE INFOCOM 2023. May 17–20, 2023. New York, NY, USA.  
(Acceptance rate: 19.2%; BK21 IF=4; Top conference in computer networks)
- [c4] WebMythBusters: An In-depth Study of Mobile Web Experience  
[Seonghoon Park](#), Yonghun Choi, and Hojung Cha  
IEEE INFOCOM 2021 – IEEE Conference on Computer Communications  
IEEE INFOCOM 2021. May 10–13, 2021. Virtual Conference.  
(Acceptance rate: 19.7%; BK21 IF=4; Top conference in computer networks)
- [c3] GAZEL: Runtime Gaze Tracking for Smartphones  
Joonbeom Park, [Seonghoon Park](#), and Hojung Cha  
The 19th International Conference on Pervasive Computing and Communications  
IEEE PerCom 2021. March 22–26, 2021. Virtual Conference.  
(Acceptance rate: 10.6% for full papers; BK21 IF=3)
- [c2] Optimizing Energy Efficiency of Browsers in Energy-Aware Scheduling-enabled Mobile Devices  
Yonghun Choi, [Seonghoon Park](#), and Hojung Cha  
The 25th Annual International Conference on Mobile Computing and Networking  
ACM MobiCom 2019. October 21–25, 2019. Los Cabos, Mexico.  
(Acceptance rate: 19.0%; BK21 IF=4; Top conference in mobile computing)
- [c1] Graphics-aware Power Governing for Mobile Devices  
Yonghun Choi, [Seonghoon Park](#), and Hojung Cha  
The 17th Annual International Conference on Mobile Systems, Applications, and Services  
ACM MobiSys 2019. June 17–21, 2019. Seoul, South Korea.  
(Acceptance rate: 22.7%; BK21 IF=3; Top conference in mobile systems)

## Journal Papers

- [j2] Duration-Aware Sound Event Detection on Ultra-Low-Power Sensor Devices  
[Seonghoon Park](#), Junick Ahn, Daeyong Kim, and Hojung Cha  
ACM Transactions on Embedded Computing Systems, Accepted.  
(JCR 2024 IF=2.6, Q2)
- [j1] Optimizing Energy Consumption of Mobile Games  
Yonghun Choi, [Seonghoon Park](#), Seunghyeok Jeon, and Hojung Cha  
IEEE Transactions on Mobile Computing, Vol. 21, Issue 10, Oct. 2022, pp 3744–3756.  
(JCR 2023 IF=7.7; Top 5% journal in Computer Science, Information Systems)

## Under Review/Revision

[u5] Anonymized Paper (On-Device AI)

Co-author

Under Review (Conference)

[u4] Anonymized Paper (Assistive web interaction)

Co-author

Minor Review (Journal)

[u3] Anonymized Paper (Mobile immersive computing)

First author

Under Review (Conference)

[u2] Anonymized Paper (Energy harvesting, Mobile immersive computing)

Co-first author

Under Review (Journal)

[u1] Anonymized Paper (On-device AI)

Co-author

Under Review (Conference)

## PATENTS

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[p3] I/O Sharing Device and Method

[Seonghoon Park](#), Jeho Lee, and Hojung Cha

Patent No. 10-2823808 (Republic of Korea; granted Jun. 18, 2025)

[p2] "Method for Omnidirectional 3D Object Detection, Program Performing the Method, and Computing Device Executing the Program"

Jeho Lee, Chanyoung Jung, [Seonghoon Park](#), Hyungchol Jun, and Hojung Cha

Patent Pending, Patent Application No. 10-2024-0120347 (Republic of Korea; filed Sep. 04, 2024)

[p1] "System and Operating Method for Cross-Device Experiences using In-Browser Virtual Proxy"

[Seonghoon Park](#), Jeho Lee, and Hojung Cha

Patent Pending, Patent Application No. 10-2024-0112156 (Republic of Korea; filed Aug. 21, 2024)

## RESEARCH PROJECTS

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Development of AI-powered Real-time Cross-device 360-degree Video Sharing Technique

RS-2024-00412632, NRF, Republic of Korea

Sep. 2024 – Aug. 2025

Development of On-device DNN Inference System for Real-time 3D Perception with Mobile 360-degree Camera

RS-2024-00344323, NRF, Republic of Korea

May. 2024 – Aug. 2025

Development of High-Assurance ( $\geq$ EAL6) Secure Microkernel

RS-2018-II180532, IITP, Republic of Korea

Apr. 2018 – Aug. 2025

Task Relation Graph Prediction based on RNN  
Samsung Electronics, Republic of Korea

*Mar. 2023 – Feb. 2024*

Development of Energy Management Techniques for Batteryless IoT System  
2019R1A2C200461913, NRF, Republic of Korea

*Mar. 2019 – Feb. 2022*

Highly Flexible Device Profiling and Analysis System for Web Experiences Measurement  
2017M3C4A708367723, NRF, Republic of Korea

*Nov. 2017 – Dec. 2020*

System Software for Mobile Device Power Management to Improve Available Time by 30%  
Samsung Science & Technology Foundation, Samsung Electronics, Republic of Korea

*Jan. 2017 – Aug. 2018*

## ACADEMIC SERVICES

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### Technical Program Committee

- ACM MobiSys: 2026 (Light)

### Peer Reviewer

- IEEE Transactions on Mobile Computing
- IEEE/ACM Transactions on Networking

## TEACHING EXPERIENCES

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### System Programming (CSI 3107)

Teaching Assistant—Yonsei University, Seoul, Republic of Korea

*Fall 2024, Fall 2020, Fall 2019, Fall 2018*

### Operating Systems (CSI3101)

Teaching Assistant—Yonsei University, Seoul, Republic of Korea

*Spring 2020, Spring 2019, Spring 2018*

## AWARDS AND HONORS

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### Best Paper Award

ACM MobiSys 2025

*Jun. 2025*

### Ph.D. Fellowship

National Research Foundation of Korea (NRF), Republic of Korea

*Sep. 2024 – Aug. 2025*

### Honors

Department of Computer Science,  
Yonsei University, Seoul, Republic of Korea

*2017 Fall, 2017 Spring, 2014 Fall, 2014 Spring*

## TECHNICAL SKILLS

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### Language

- Korean (Native)
- English

### Programming Skills

- Machine learning frameworks (PyTorch, TensorFlow, TensorFlow Lite)
- Android applications
- Operating systems (Android kernel, Android framework, ChibiOS/RT microkernel)
- Web programming (Web applications with Node.js, Flask, etc.; Web extensions)